

Лабораторная работа №6

Дисциплина: Основы информационной безопасности

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Цель работы

Развить навыки администрирования ОС linux. Получить практическое знакомство с SELinux1. Проверить работу SELinux на практике совместно с веб-сервером Apache.

Выполнение лабораторной работы

Вошел в систему под своей учетной записью. Убедилась, что SELinux работает в режиме enforcing политики targeted с помощью getenforce и sestatus

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Memory: 13.6M (peak: 13.9M)
CPU: 97ms
CGroup: /system.slice/httpd.service
├─7130 /usr/sbin/httpd -DFOREGROUND
├─7131 /usr/sbin/httpd -DFOREGROUND
├─7132 /usr/sbin/httpd -DFOREGROUND
├─7134 /usr/sbin/httpd -DFOREGROUND
└─7184 /usr/sbin/httpd -DFOREGROUND

Aug 29 21:49:52 localhost.localdomain systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 29 21:49:52 localhost.localdomain (httpd)[7130]: httpd.service: Referenced but unset environment variable eval
Aug 29 21:49:52 localhost.localdomain httpd[7130]: AH00558: httpd: Could not reliably determine the server's fully
Aug 29 21:49:52 localhost.localdomain systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 29 21:49:52 localhost.localdomain httpd[7130]: Server configured, listening on: port 80
~
root@localhost:~# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.servic
e'.
root@localhost:~# iptables -F
root@localhost:~# iptables -P INPUT ACCEPT
root@localhost:~# iptables -P OUTPUT ACCEPT
root@localhost:~# nano /etc/httpd/conf/httpd.conf
root@localhost:~# systemctl restart httpd
root@localhost:~# getenforce
Enforcing
root@localhost:~# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:       /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:        enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     33
root@localhost:~#
```

Рисунок 1. проверка режима работы SELinux

Запускаю сервер apache, он работает, что видно из вывода команды service httpd status

```
root@localhost:~# sudo -i
guest@localhost:~# sudo -i
guest2@localhost:~#
root@localhost:~# sudo -i

root@localhost:~# getenforce
Enforcing
root@localhost:~# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:      /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:        enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     33
root@localhost:~# service httpd status
Redirecting to /bin/systemctl status httpd.service
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-29 21:54:02 CEST; 2min 8s ago
 Invocation: 7ae5621567ff4a338b0a16abda605ebd
    Docs: man:httpd.service(8)
   Main PID: 7518 (httpd)
      Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 12175)
    Memory: 14.1M (peak: 14.4M)
       CPU: 182ms
    CGroup: /system.slice/httpd.service
            └─7518 /usr/sbin/httpd -DFOREGROUND
              └─7519 /usr/sbin/httpd -DFOREGROUND
                └─7520 /usr/sbin/httpd -DFOREGROUND
                  └─7521 /usr/sbin/httpd -DFOREGROUND
                    └─7529 /usr/sbin/httpd -DFOREGROUND

Aug 29 21:54:02 localhost.localdomain systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 29 21:54:02 localhost.localdomain (httpd)[7518]: httpd.service: Referenced but unset environment variable eval
Aug 29 21:54:02 localhost.localdomain httpd[7518]: Server configured, listening on: port 80
Aug 29 21:54:02 localhost.localdomain systemd[1]: Started httpd.service - The Apache HTTP Server.
lines 1-21/21 (END)
```

Рисунок 2. Проверка работы Apache

С помощью команды `ps auxZ | grep httpd` нашел веб-сервер Apache в списке процессов. Его контекст безопасности - `httpd_t`

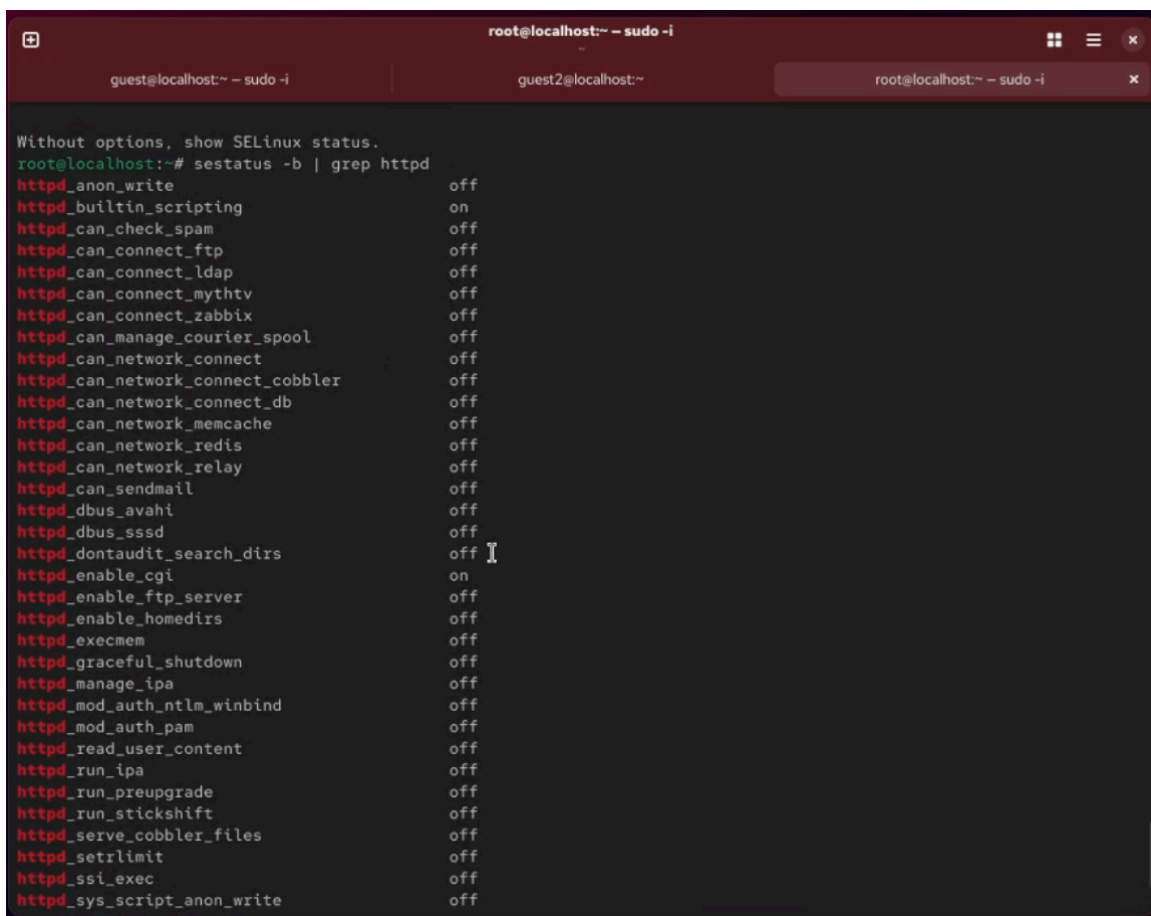
```
root@localhost:~# sudo -i
guest@localhost:~# sudo -i
guest2@localhost:~#
root@localhost:~# sudo -i

root@localhost:~# service httpd status
Redirecting to /bin/systemctl status httpd.service
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-29 21:54:02 CEST; 2min 8s ago
 Invocation: cae5621567ff4a338b0a16abda605ebd
    Docs: man:httpd.service(8)
  Main PID: 7518 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 12175)
   Memory: 14.1M (peak: 14.4M)
     CPU: 182ms
    CGroup: /system.slice/httpd.service
            └─7518 /usr/sbin/httpd -DFOREGROUND
              └─7519 /usr/sbin/httpd -DFOREGROUND
                └─7520 /usr/sbin/httpd -DFOREGROUND
                  └─7521 /usr/sbin/httpd -DFOREGROUND
                    └─7529 /usr/sbin/httpd -DFOREGROUND

Aug 29 21:54:02 localhost.localdomain systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 29 21:54:02 localhost.localdomain (httpd)[7518]: httpd.service: Referenced but unset environment variable eval
Aug 29 21:54:02 localhost.localdomain httpd[7518]: Server configured, listening on: port 80
Aug 29 21:54:02 localhost.localdomain systemd[1]: Started httpd.service - The Apache HTTP Server.
root@localhost:~# ps auxZ | grep httpd
system_u:system_r:httpd_t:s0 root 7518 0.0 0.5 18836 10884 ? Ss 21:54 0:00 /usr/sbin/httpd
-DFOREGROUND
system_u:system_r:httpd_t:s0 apache 7519 0.0 0.2 18680 5068 ? S 21:54 0:00 /usr/sbin/httpd
-DFOREGROUND
system_u:system_r:httpd_t:s0 apache 7520 0.0 0.3 1436720 7612 ? Sl 21:54 0:00 /usr/sbin/httpd
-DFOREGROUND
system_u:system_r:httpd_t:s0 apache 7521 0.0 0.3 1567856 7848 ? Sl 21:54 0:00 /usr/sbin/httpd
-DFOREGROUND
system_u:system_r:httpd_t:s0 apache 7529 0.0 0.3 1436720 7612 ? Sl 21:54 0:00 /usr/sbin/httpd
-DFOREGROUND
unconfined_u:unconfined_r:unconfined_t:s0-c0.c1023 root 7731 0.0 0.1 227700 2136 pts/4 S+ 21:56 0:00 grep --c
olor=auto httpd
root@localhost:~#
```

Рисунок 3. Контекст безопасности Apache

Просмотрел текущее состояние переключателей SELinux для Apache

A terminal window titled 'root@localhost:~ - sudo -i' with three tabs: 'guest@localhost:~ - sudo -i', 'guest2@localhost:~', and 'root@localhost:~ - sudo -i'. The terminal shows the command 'sestatus -b | grep httpd' and its output, which lists various SELinux booleans for the httpd process and their current states. The output is as follows:

```
Without options, show SELinux status.
root@localhost:~# sestatus -b | grep httpd
httpd_anon_write                off
httpd_builtin_scripting         on
httpd_can_check_spam            off
httpd_can_connect_ftp           off
httpd_can_connect_ldap          off
httpd_can_connect_mythtv        off
httpd_can_connect_zabbix        off
httpd_can_manage_courier_spool   off
httpd_can_network_connect       off
httpd_can_network_connect_cobbler off
httpd_can_network_connect_db     off
httpd_can_network_memcache       off
httpd_can_network_redis          off
httpd_can_network_relay          off
httpd_can_sendmail              off
httpd_dbus_avahi                 off
httpd_dbus_sssd                  off
httpd_dontaudit_search_dirs      off
httpd_enable_cgi                 on
httpd_enable_ftp_server          off
httpd_enable_homedirs            off
httpd_execmem                    off
httpd_graceful_shutdown          off
httpd_manage_ipa                 off
httpd_mod_auth_ntlm_winbind      off
httpd_mod_auth_pam               off
httpd_read_user_content          off
httpd_run_ipa                    off
httpd_run_preupgrade             off
httpd_run_stickshift             off
httpd_serve_cobbler_files        off
httpd_setrlimit                  off
httpd_ssi_exec                   off
httpd_sys_script_anon_write      off
```

Рисунок 4. Состояние переключателей SELinux

Просмотрел статистику по политике с помощью команды seinfo. Множество пользователей - 8, ролей - 15, типов - 4718.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

setools-console-4.6.0-3.el10.x86_64 Policy analysis command-line tools for SELinux
Proceed with changes? [N/y] y

* Waiting in queue...
* Waiting for authentication...
* Waiting in queue...
* Downloading packages...
* Requesting data...
* Testing changes...
* Installing packages...
Statistics for policy file: /sys/fs/selinux/policy
Policy Version: 33 (MLS enabled)
Target Policy: selinux
Handle unknown classes: allow
Classes: 133 Permissions: 458
Sensitivities: 1 Categories: 1024
Types: 4718 Attributes: 251
Users: 8 Roles: 15
Booleans: 328 Cond. Expr.: 356
Allow: 59835 Neverallow: 0
Auditallow: 156 Dontaudit: 8001
Type_trans: 244193 Type_change: 80
Type_member: 34 Range_trans: 3893
Role allow: 39 Role_trans: 318
Constraints: 70 Validatetrans: 0
MLS Constrains: 72 MLS Val. Tran: 0
Permissives: 23 Polcap: 6
Defaults: 7 Typebounds: 0
Allowxperm: 0 Neverallowxperm: 0
Auditallowxperm: 0 Dontauditxperm: 0
Ibendportcon: 0 Ibpkeycon: 0
Initial SIDs: 27 Fs_use: 35
Genfscon: 112 Portcon: 667
Netifcon: 0 Nodecon: 0

root@localhost:~#
```

Рисунок 5. Статистика по политике

Типы поддиректорий, находящихся в директории /var/www, с помощью команды `ls -lZ /var/www` следующие: владелец - root, права на изменения только у владельца. Файлов в директории нет

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Allowxperm:      0   Neverallowxperm:    0
Auditallowxperm: 0   Dontauditxperm:    0
Ibendportcon:    0   Ibpkeycon:         0
Initial SIDs:    27   Fs_use:            35
Genfscon:        112  Portcon:           667
Netifcon:        0   Nodecon:           0

root@localhost:~# seinfo
Statistics for policy file: /sys/fs/selinux/policy
Policy Version:      33 (MLS enabled)
Target Policy:       selinux
Handle unknown classes: allow
Classes:             133   Permissions:        458
Sensitivities:       1    Categories:         1024
Types:               4718  Attributes:         251
Users:               8    Roles:              15
Booleans:            328   Cond. Expr.:        356
Allow:               59835 Neverallow:          0
Auditallow:          156   Dontaudit:           8001
Type_trans:          244193 Type_change:         80
Type_member:         34    Range_trans:        3893
Role allow:          39    Role_trans:         318
Constraints:         70    Validatetrans:       0
MLS Constrain:       72    MLS Val. Tran:       0
Permissives:        23    Polcap:              6
Defaults:            7    Typebounds:          0
Allowxperm:          0     Neverallowxperm:    0
Auditallowxperm:     0     Dontauditxperm:    0
Ibendportcon:        0     Ibpkeycon:         0
Initial SIDs:        27     Fs_use:            35
Genfscon:            112    Portcon:           667
Netifcon:            0     Nodecon:           0

root@localhost:~# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 20 02:00 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Aug 20 02:00 html
root@localhost:~#
```

Рисунок 6. Типы поддиректорий

В директории /var/www/html нет файлов.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Ibendportcon:      0    Ibpkeycon:      0
Initial SIDs:      27    Fs_use:         35
Genfscon:          112   Portcon:        667
Netifcon:          0    Nodecon:        0

root@localhost:~# seinfo
Statistics for policy file: /sys/fs/selinux/policy
Policy Version:    33 (MLS enabled)
Target Policy:     selinux
Handle unknown classes: allow
Classes:           133   Permissions:      458
Sensitivities:     1    Categories:       1024
Types:             4718  Attributes:       251
Users:             8     Roles:            15
Booleans:          328   Cond. Expr.:     356
Allow:             59835 Neverallow:        0
Auditallow:        156   Dontaudit:        8001
Type_trans:        244193 Type_change:       80
Type_member:        34   Range_trans:      3893
Role allow:         39   Role_trans:       318
Constraints:        70   Validatetrans:    0
MLS Constrain:      72   MLS Val. Tran:    0
Permissives:        23   Polcap:           6
Defaults:           7    Typebounds:       0
Allowxperm:         0    Neverallowxperm:  0
Auditallowxperm:    0    Dontauditxperm:   0
Ibendportcon:      0    Ibpkeycon:        0
Initial SIDs:      27    Fs_use:           35
Genfscon:          112   Portcon:          667
Netifcon:          0    Nodecon:          0

root@localhost:~# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 20 02:00 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Aug 20 02:00 html
root@localhost:~# ls -lZ /var/www/html
total 0
root@localhost:~#
```

Рисунок 7. Типы файлов

Создать файл может только суперпользователь, поэтому от его имени создаем файл touch.html со следующим содержанием:

```
<html>
<body>test</body>
</html>
```

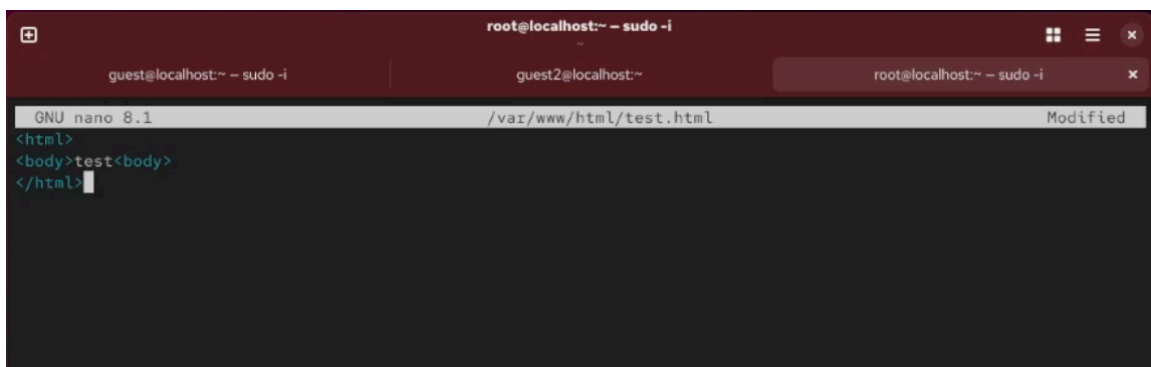



Рисунок 8. Создание файла

Проверяю контекст созданного файла. По умолчанию это httpd_sys_content_t.

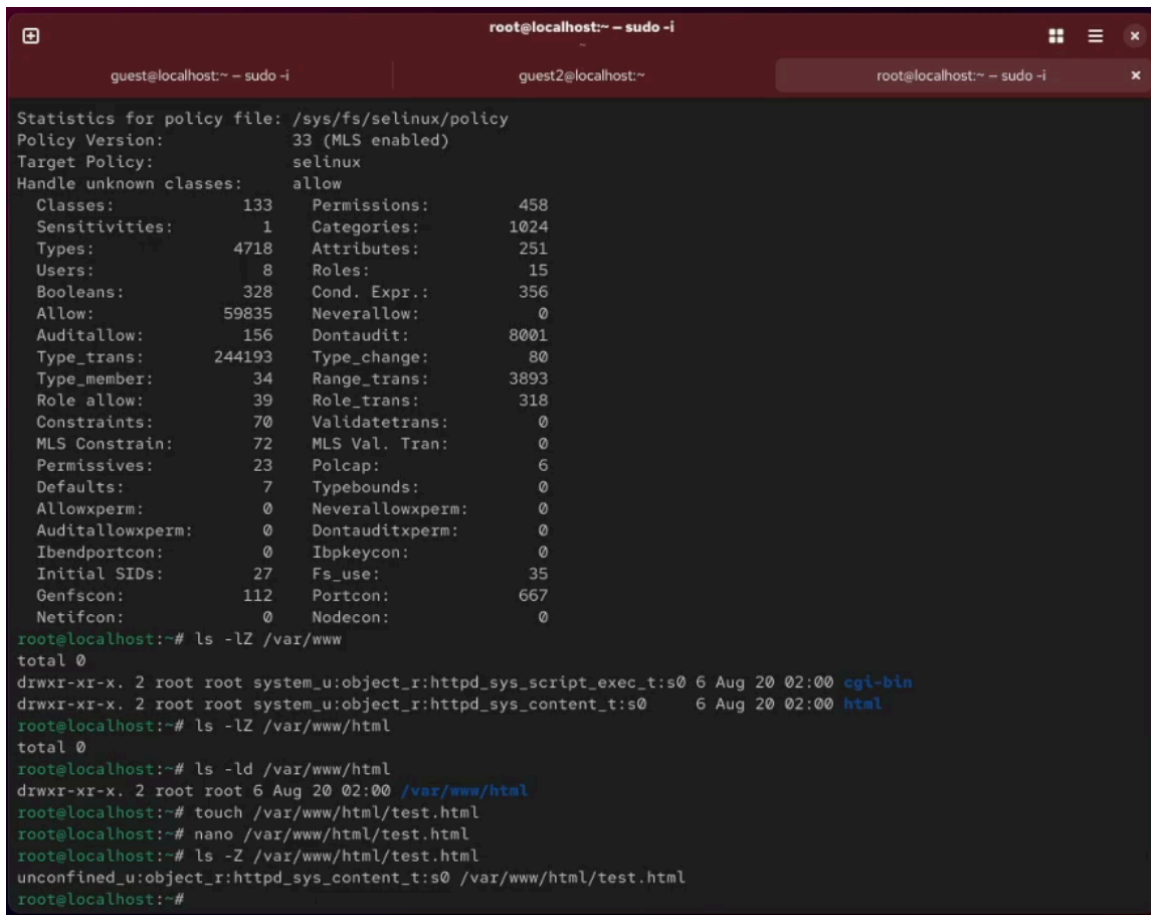


Рисунок 9. Контекст файла

Обращаюсь к файлу через веб-сервер, введя в браузере адрес <http://127.0.0.1/test.html>. Файл был успешно отображён.

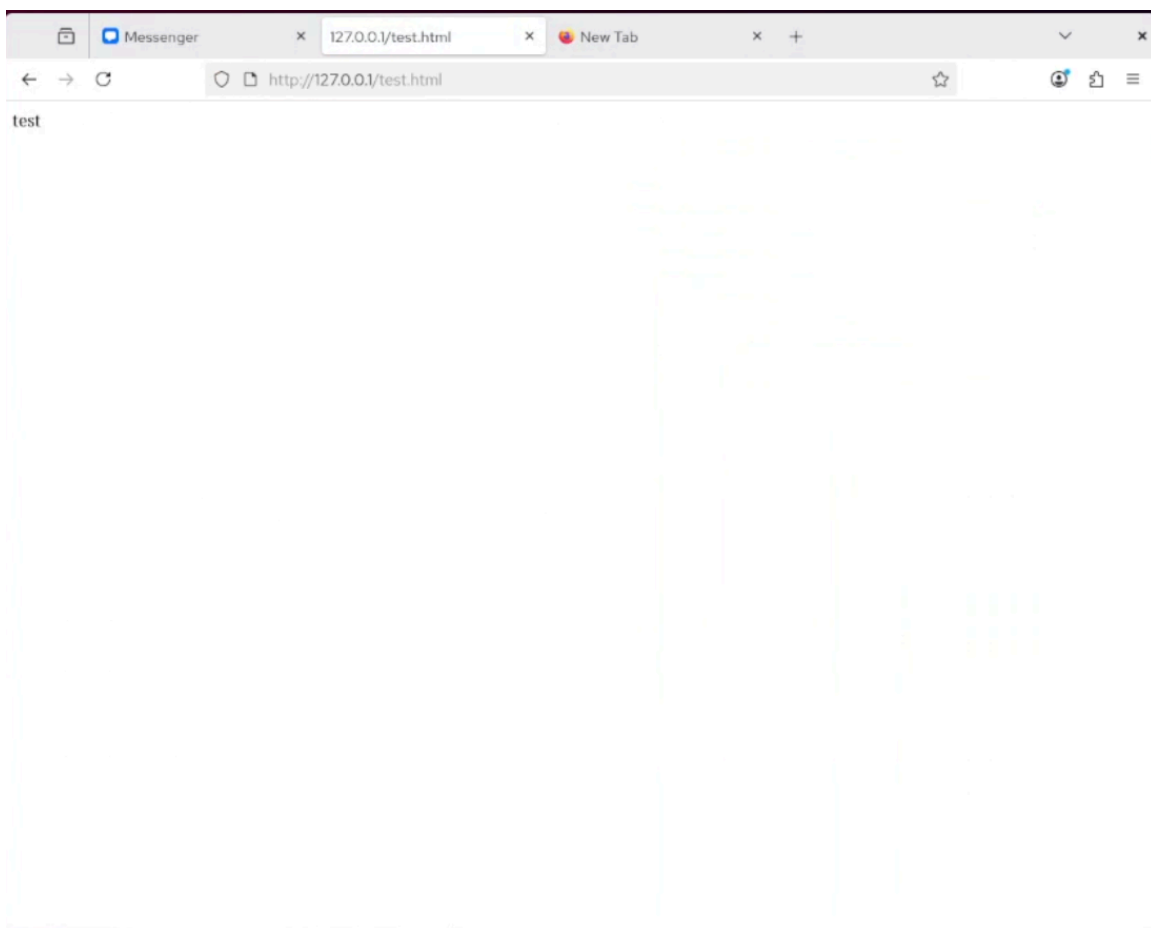


Рисунок 10. Отображение файла

Изменяю контекст файла test.html с httpd_sys_content_t на любой другой, к которому процесс httpd не должен иметь доступа, например, на samba_share_t: `chcon -t samba_share_t /var/www/html/test.html`. Контекст действительно поменялся.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Type_member:      34    Range_trans:      3893
Role_allow:       39    Role_trans:       318
Constraints:      70    Validatetrans:    0
MLS Constrain:    72    MLS Val. Tran:    0
Permissives:      23    Polcap:           6
Defaults:         7    Typebounds:       0
Allowxperm:       0    Neverallowxperm:  0
Auditallowxperm:  0    Dontauditxperm:   0
Ibendportcon:     0    Ibkeycon:         0
Initial SIDs:     27    Fs_use:           35
Genfscon:         112   Portcon:          667
Netifcon:         0    Nodecon:          0

root@localhost:~# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 20 02:00 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Aug 20 02:00 html
root@localhost:~# ls -lZ /var/www/html
total 0
root@localhost:~# ls -ld /var/www/html
drwxr-xr-x. 2 root root 6 Aug 20 02:00 /var/www/html
root@localhost:~# touch /var/www/html/test.html
root@localhost:~# nano /var/www/html/test.html
root@localhost:~# ls -Z /var/www/html/test.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/test.html
root@localhost:~# curl http://127.0.0.1/test.html
<html>
<body>test<body>
</html>
root@localhost:~# man httpd_selinux
No manual entry for httpd_selinux
root@localhost:~# chcon -t samba_share_t /var/www/html/test.html
root@localhost:~# ls -z /var/www/html/test.html
ls: invalid option -- 'z'
Try 'ls --help' for more information.
root@localhost:~# ls -Z /var/www/html/test.html
unconfined_u:object_r:samba_share_t:s0 /var/www/html/test.html
root@localhost:~#
```

Рисунок 11. Изменение контекста

При попытке отображения файла получаем сообщение об ошибке 404.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Ibendportcon:      0      Ibpkeycon:      0
Initial SIDs:      27      Fs_use:      35
Genfscon:      112      Portcon:      667
Netifcon:      0      Nodecon:      0

root@localhost:~# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 20 02:00 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Aug 20 02:00 html
root@localhost:~# ls -lZ /var/www/html
total 0
root@localhost:~# ls -ld /var/www/html
drwxr-xr-x. 2 root root 6 Aug 20 02:00 /var/www/html
root@localhost:~# touch /var/www/html/test.html
root@localhost:~# nano /var/www/html/test.html
root@localhost:~# ls -Z /var/www/html/test.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/test.html
root@localhost:~# curl http://127.0.0.1/test.html
<html>
<body>test<body>
</html>
root@localhost:~# man httpd_selinux
No manual entry for httpd_selinux
root@localhost:~# chcon -t samba_share_t /var/www/html/test.html
root@localhost:~# ls -z /var/www/html/test.html
ls: invalid option -- 'z'
Try 'ls --help' for more information.
root@localhost:~# ls -Z /var/www/html/test.html
unconfined_u:object_r:samba_share_t:s0 /var/www/html/test.html
root@localhost:~# curl http://127.0.0.1/test.html
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
```

Рисунок 12. Отображение файла

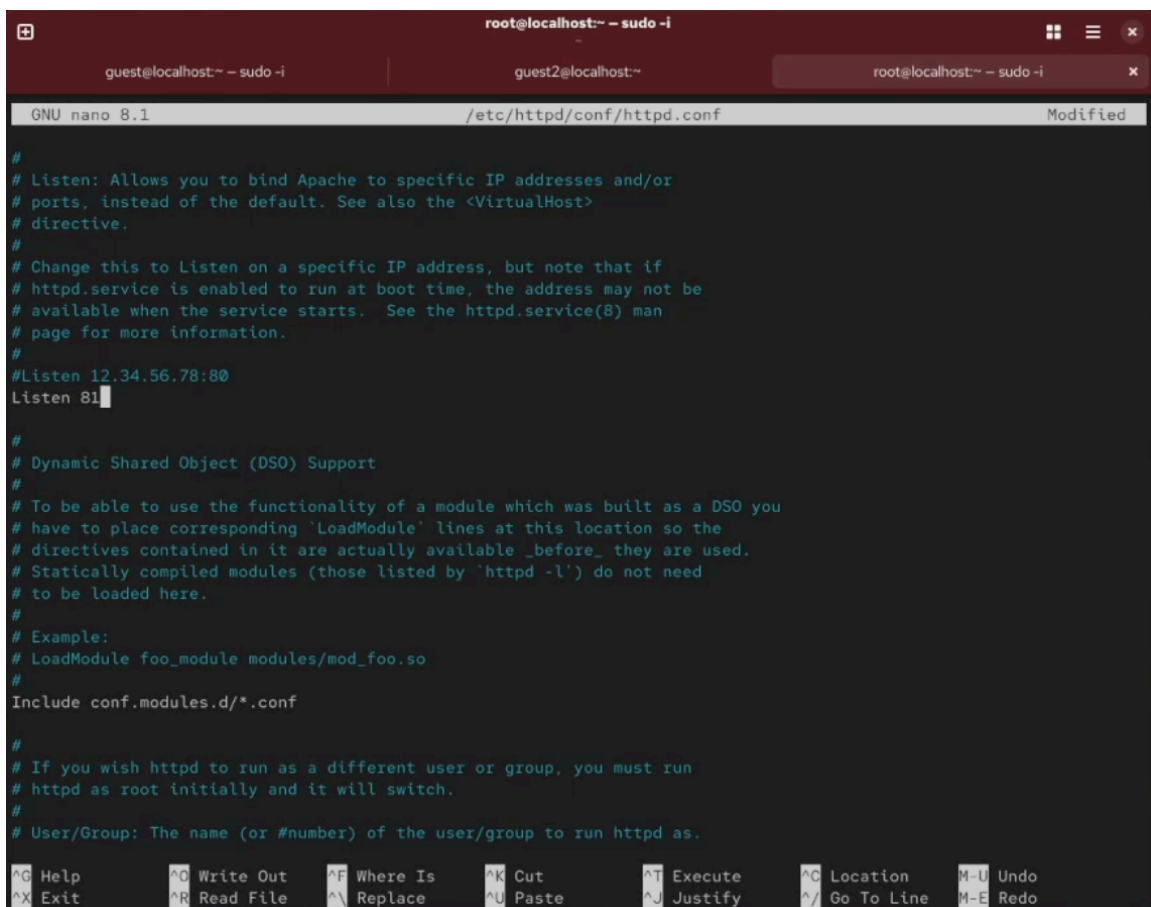
файл не был отображён, хотя права доступа позволяют читать этот файл любому пользователю, потому что установлен контекст, к которому процесс httpd не должен иметь доступа. Просматриваю log-файлы веб-сервера Apache и системный лог-файл: tail /var/log/messages.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

le /var/www/html/test.html. For complete SELinux messages run: sealert -l cf6b9508-4f57-4eee-a2bc-49e6f588a4b7
Aug 29 22:11:35 localhost setroubleshoot[8018]: SELinux is preventing /usr/sbin/httpd from getattr access on the fi
le /var/www/html/test.html.#012#012***** Plugin restorecon (92.2 confidence) suggests *****#0
12#012If you want to fix the label. #012/var/www/html/test.html default label should be httpd_sys_content_t.#012The
n you can run restorecon. The access attempt may have been stopped due to insufficient permissions to access a pare
nt directory in which case try to change the following command accordingly.#012Do#012# /sbin/restorecon -v /var/www
/html/test.html#012#012***** Plugin public_content (7.83 confidence) suggests *****#012#012If you
want to treat test.html as public content#012Then you need to change the label on test.html to public_content_t or
public_content_rw_t.#012Do#012# semanage fcontext -a -t public_content_t '/var/www/html/test.html'#012# restorecon
-v '/var/www/html/test.html'#012#012***** Plugin catchall (1.41 confidence) suggests *****
#012#012If you believe that httpd should be allowed getattr access on the test.html file by default.#012Then you sh
ould report this as a bug.#012You can generate a local policy module to allow this access.#012Do#012allow this acce
ss for now by executing:#012# ausearch -c 'httpd' --raw | audit2allow -M my-httpd#012# semodule -X 300 -i my-httpd.
pp#012
Aug 29 22:11:35 localhost setroubleshoot[8018]: SELinux is preventing /usr/sbin/httpd from getattr access on the fi
le /var/www/html/test.html. For complete SELinux messages run: sealert -l cf6b9508-4f57-4eee-a2bc-49e6f588a4b7
Aug 29 22:11:35 localhost setroubleshoot[8018]: SELinux is preventing /usr/sbin/httpd from getattr access on the fi
le /var/www/html/test.html.#012#012***** Plugin restorecon (92.2 confidence) suggests *****#0
12#012If you want to fix the label. #012/var/www/html/test.html default label should be httpd_sys_content_t.#012The
n you can run restorecon. The access attempt may have been stopped due to insufficient permissions to access a pare
nt directory in which case try to change the following command accordingly.#012Do#012# /sbin/restorecon -v /var/www
/html/test.html#012#012***** Plugin public_content (7.83 confidence) suggests *****#012#012If you
want to treat test.html as public content#012Then you need to change the label on test.html to public_content_t or
public_content_rw_t.#012Do#012# semanage fcontext -a -t public_content_t '/var/www/html/test.html'#012# restorecon
-v '/var/www/html/test.html'#012#012***** Plugin catchall (1.41 confidence) suggests *****
#012#012If you believe that httpd should be allowed getattr access on the test.html file by default.#012Then you sh
ould report this as a bug.#012You can generate a local policy module to allow this access.#012Do#012allow this acce
ss for now by executing:#012# ausearch -c 'httpd' --raw | audit2allow -M my-httpd#012# semodule -X 300 -i my-httpd.
pp#012
Aug 29 22:11:45 localhost systemd[1]: dbus-1.12.0-1.fc28.fedora.project.SetroubleshootPrivileged@1.service: Deactivated s
uccessfully.
Aug 29 22:11:45 localhost systemd[1]: setroubleshootd.service: Deactivated successfully.
Aug 29 22:11:45 localhost systemd[1]: setroubleshootd.service: Consumed 459ms CPU time, 68.8M memory peak.
Aug 29 22:16:27 localhost systemd[1]: Starting fwupd-refresh.service - Refresh fwupd metadata and update motd...
Aug 29 22:16:27 localhost systemd[1]: fwupd-refresh.service: Deactivated successfully.
Aug 29 22:16:27 localhost systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metadata and update motd.
root@localhost:~#
```

Рисунок 13. Попытка прочесть лог-файл

Чтобы запустить веб-сервер Apache на прослушивание TCP-порта 81 (а не 80, как рекомен-
дует IANA и прописано в /etc/services) открываю файл /etc/httpd/httpd.conf для изменения.
Нахожу строчку Listen 80 и заменяю её на Listen 81.



```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

GNU nano 8.1 /etc/httpd/conf/httpd.conf Modified

#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 81

#
# Dynamic Shared Object (DSO) Support
#
# To be able to use the functionality of a module which was built as a DSO you
# have to place corresponding 'LoadModule' lines at this location so the
# directives contained in it are actually available _before_ they are used.
# Statically compiled modules (those listed by 'httpd -l') do not need
# to be loaded here.
#
# Example:
# LoadModule foo_module modules/mod_foo.so
#
Include conf.modules.d/*.conf

#
# If you wish httpd to run as a different user or group, you must run
# httpd as root initially and it will switch.
#
# User/Group: The name (or #number) of the user/group to run httpd as.

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Рисунок 14. Изменение файла

Выполняю команду `semanage port -a -t http_port_t -p tcp 81`. После этого проверяю список портов командой `semanage port -l | grep http_port_t`. Порт 81 появился в списке

```
root@localhost:~# sudo -i
guest@localhost:~# sudo -i
guest2@localhost:~#
root@localhost:~# sudo -i

Aug 29 22:11:35 localhost setroubleshoot[8018]: SELinux is preventing /usr/sbin/httpd from getattr access on the file /var/www/html/test.html. #012#012***** Plugin restorecon (92.2 confidence) suggests *****#012#012If you want to fix the label. #012/var/www/html/test.html default label should be httpd_sys_content_t.#012Then you can run restorecon. The access attempt may have been stopped due to insufficient permissions to access a parent directory in which case try to change the following command accordingly.#012Do#012# /sbin/restorecon -v /var/www/html/test.html#012#012***** Plugin public_content (7.83 confidence) suggests *****#012#012If you want to treat test.html as public content#012Then you need to change the label on test.html to public_content_t or public_content_rw_t.#012Do#012# semanage fcontext -a -t public_content_t '/var/www/html/test.html'#012# restorecon -v '/var/www/html/test.html'#012#012***** Plugin catchall (1.41 confidence) suggests *****#012#012If you believe that httpd should be allowed getattr access on the test.html file by default.#012Then you should report this as a bug.#012You can generate a local policy module to allow this access.#012Do#012allow this access for now by executing:#012# ausearch -c 'httpd' --raw | audit2allow -M my-httpd#012# semodule -X 300 -i my-httpd.pp#012
Aug 29 22:11:45 localhost systemd[1]: dbus-1.1-org.fedoraproject.SetroubleshootPrivileged@1.service: Deactivated successfully.
Aug 29 22:11:45 localhost systemd[1]: setroubleshootd.service: Deactivated successfully.
Aug 29 22:11:45 localhost systemd[1]: setroubleshootd.service: Consumed 459ms CPU time, 68.8M memory peak.
Aug 29 22:16:27 localhost systemd[1]: Starting fwupd-refresh.service - Refresh fwupd metadata and update motd...
Aug 29 22:16:27 localhost systemd[1]: fwupd-refresh.service: Deactivated successfully.
Aug 29 22:16:27 localhost systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metadata and update motd.
root@localhost:~# nano /etc/httpd.conf
root@localhost:~# nano /etc/httpd/httpd.conf
root@localhost:~# nano /etc/httpd/conf/
root@localhost:~# nano /etc/httpd/conf/httpd.conf
root@localhost:~# systemctl restart httpd
root@localhost:~# semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
root@localhost:~# semanage port -l | grep http_port_t
http_port_t          tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
http_port_t          udp      80, 443
pegasus_http_port_t  tcp      5988
root@localhost:~#
```

Рисунок 15. Проверка портов

Перезапускаю сервер Apache

```
root@localhost:~# sudo -i
guest@localhost:~# sudo -i
guest2@localhost:~#
root@localhost:~# sudo -i

root@localhost:~# nano /etc/httpd/httpd.conf
root@localhost:~# nano /etc/httpd/conf/
root@localhost:~# nano /etc/httpd/conf/httpd.conf
root@localhost:~# systemctl restart httpd
root@localhost:~# semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
root@localhost:~# semanage port -l | grep http_port_t
http_port_t          tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
http_port_t          udp      80, 443
pegasus_http_port_t  tcp      5988
root@localhost:~# curl http://127.0.0.1/test.html
curl: (7) Failed to connect to 127.0.0.1 port 80 after 0 ms: Could not connect to server
root@localhost:~# systemctl restart httpd
```

Рисунок 16. Перезапуск сервера

Теперь он работает, ведь мы внесли порт 81 в список портов httpd_port_t. Возвращаю в файле /etc/httpd/httpd.conf порт 80, вместо 81. Проверяю, что порт 81 удален, это правда.

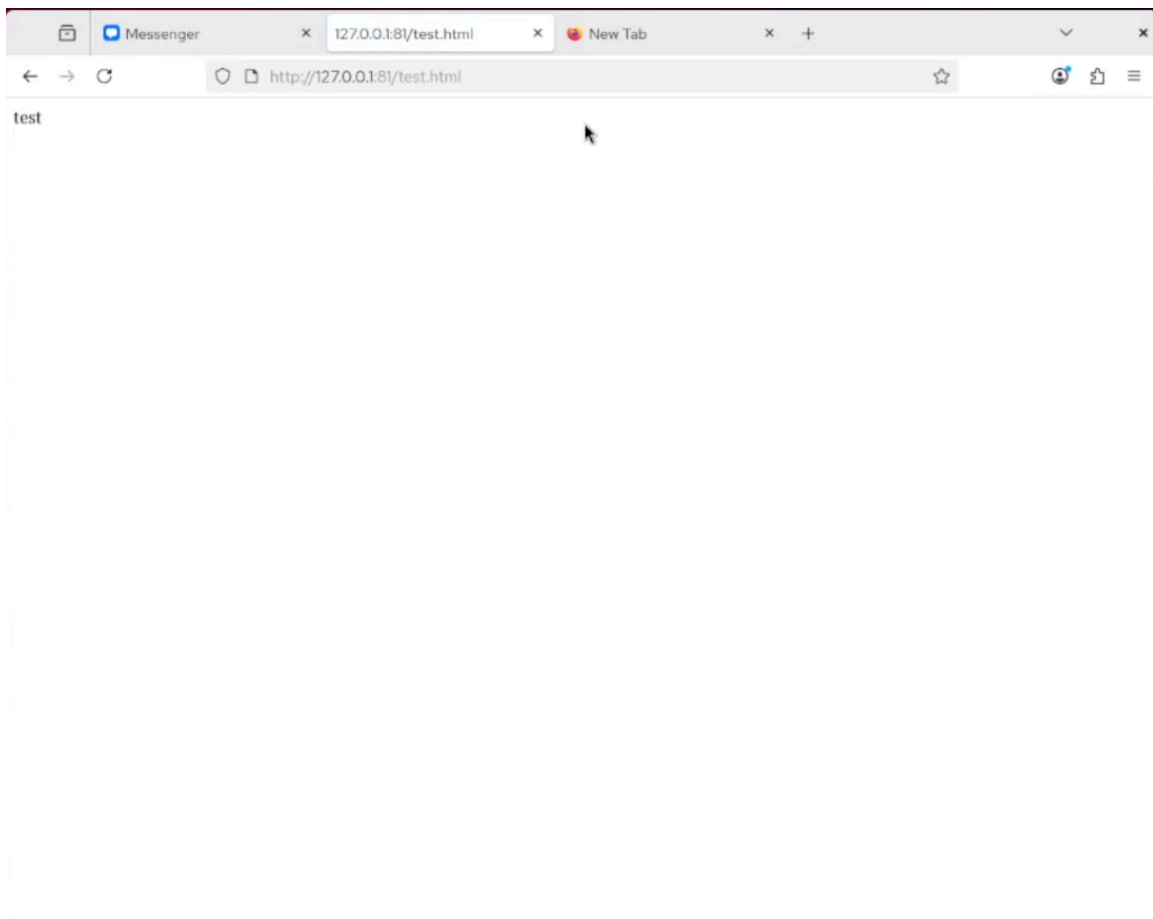
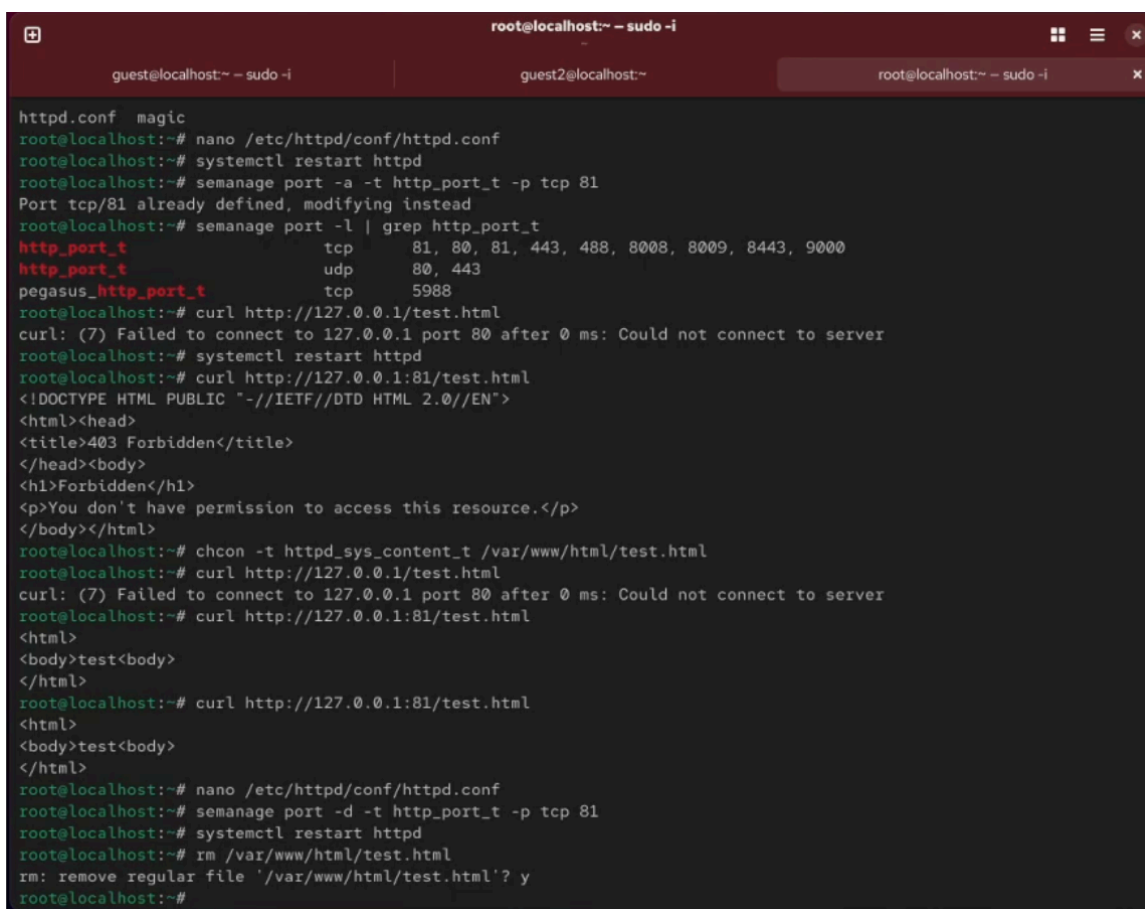


Рисунок 17. Проверка порта 81

Далее удаляю файл test.html, проверяю, что он удален

A terminal window with three tabs: 'guest@localhost:~ - sudo -i', 'guest2@localhost:~', and 'root@localhost:~ - sudo -i'. The active tab is 'root@localhost:~ - sudo -i'. The terminal shows the following commands and output:

```
httpd.conf magic
root@localhost:~# nano /etc/httpd/conf/httpd.conf
root@localhost:~# systemctl restart httpd
root@localhost:~# semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
root@localhost:~# semanage port -l | grep http_port_t
http_port_t          tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
http_port_t          udp      80, 443
pegasus_http_port_t  tcp      5988
root@localhost:~# curl http://127.0.0.1/test.html
curl: (7) Failed to connect to 127.0.0.1 port 80 after 0 ms: Could not connect to server
root@localhost:~# systemctl restart httpd
root@localhost:~# curl http://127.0.0.1:81/test.html
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
root@localhost:~# chcon -t httpd_sys_content_t /var/www/html/test.html
root@localhost:~# curl http://127.0.0.1/test.html
curl: (7) Failed to connect to 127.0.0.1 port 80 after 0 ms: Could not connect to server
root@localhost:~# curl http://127.0.0.1:81/test.html
<html>
<body>test<body>
</html>
root@localhost:~# curl http://127.0.0.1:81/test.html
<html>
<body>test<body>
</html>
root@localhost:~# nano /etc/httpd/conf/httpd.conf
root@localhost:~# semanage port -d -t http_port_t -p tcp 81
root@localhost:~# systemctl restart httpd
root@localhost:~# rm /var/www/html/test.html
rm: remove regular file '/var/www/html/test.html'? y
root@localhost:~#
```

Рисунок 18. Удаление файла

Выводы

При выполнении данной лабораторной работы были развиты навыки администрирования ОС Linux, получено первое практическое знакомство с технологией SELinux и проверена работа SELinux на практике совместно с веб-сервером Apache.

Список литературы

https://esystem.rudn.ru/pluginfile.php/3096715/mod_resource/content/2/006-lab_selinux.pdf